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Transactions between services in a multi-tenant architecture

Abstract

A method for facilitating transactions between tenants in in a multi-tenant architecture system is discussed. The method includes receiving a request, at a multi-tenant platform, from a first service of a first tenant of the multi-tenant platform to access a second service of a second tenant of the multi-tenant platform to perform a transaction, in which the request includes a first access token usable to authenticate the transaction with the first tenant. The method includes generating, by the multi-tenant platform using the first access token, a universal access token. The method includes generating, by the multi-tenant platform using the universal access token, a second access token useable to authenticate the transaction with the second tenant. The method includes using, by the multi-tenant platform, the second access token to communicate with the second service to perform the transaction.

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Background/Summary

(1) The present application is a continuation of U.S. patent application Ser. No. 16/457,822, filed Jun. 28, 2019, which is a continuation-in-part of U.S. application Ser. No. 16/009,994, filed Jun. 15, 2018; the disclosure of the above-referenced application is incorporated by reference herein in its entirety.

BACKGROUND

(1) Embodiments of the present disclosure generally relate to the field of software architecture and, more particularly, to managing how various entities are on-boarded, managed, and/or accessed in a multi-tenant system architecture.

(2) Complex software systems can have various architectures, including a single-tenant software architecture and a multi-tenant software architecture, among others. In a single-tenant software architecture, each entity (such as a company or a portion of that company) can have their own instances of each software application, data, and any supporting infrastructure. In a single-tenant architecture, the software for each tenant entity can be customized as desired. However, drawbacks of using the single tenant architecture include expense in resources, as well as requirements to host,

customize, and maintain separate software applications for each tenant.

(3) In contrast, in a multi-tenant software architecture, each entity can share the same instances of applications, data, and/or infrastructure. A multi-tenant software provider can provision their applications and/or resources to multiple entities from a single software architecture. In multi-tenant software architecture, access to data can be shared among the various entities. By sharing much of the applications, data, and software; various resources such as installation, configuration, number of physical servers, maintenance, and even power can be optimized. However, drawbacks of using multi-tenant architecture include complexity of managing how multiple software applications are configured and shared among multiple clients. Also, design of multi-tenant architecture-based software systems that provides secure and reliable access to the data can be problematic. Furthermore, interoperability difficulties may arise when using software applications, in a multi-tenant architecture system, that were initially managed and/or hosted by different businesses.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present embodiments may be better understood, and numerous objects, features, and advantages made apparent to those skilled in the art by referencing the accompanying drawings.

(2) FIG. 1 is a system diagram illustrating embodiments of a multi-tenant software architecture system that can be accessed by communicating with a user device.

(3) FIG. 2 is a system diagram illustrating embodiments of the multi-tenant software architecture of FIG. 1 including unified identity services.

(4) FIG. 3 is a diagram illustrating hierarchical data structures used by the multi-tenant software architecture systems that provide unified identity services.

(5) FIG. 4 is another diagram illustrating hierarchical data structures used by the multi-tenant software architecture systems that provide unified identity services.

(6) FIG. 5 is a flow diagram illustrating embodiments of operations for facilitating a transaction between services of tenants of a multi-tenant platform.

(7) FIGS. 6A and 6B are flow diagrams illustrating embodiments of operations for facilitating a transaction between services of tenants of a multi-tenant platform.

(8) FIG. 7 is a flow diagram illustrating an embodiment of a transaction method in accordance with the disclosed embodiments.

(9) FIG. 8 is a block diagram illustrating embodiments of electronic devices used in the communication systems of FIGS. 1-7.

(10) This disclosure includes references to “one embodiment” or “an embodiment.” The appearances of the phrases “in one embodiment” or “in an embodiment” do not necessarily refer to the same embodiment. Particular features, structures, or characteristics may be combined in any suitable manner consistent with this disclosure.

(11) Within this disclosure, different entities (which may variously be referred to as “units,” “circuits,” other components, etc.) may be described or claimed as “configured” to perform one or more tasks or operations. This formulation—[entity] configured to [perform one or more tasks]—is used herein to refer to structure (i.e., something physical, such as an electronic circuit). More specifically, this formulation is used to indicate that this structure is arranged to perform the one or more tasks during operation. A structure can be said to be “configured to” perform some task even if the structure is not currently being operated. A “computer system configured to receive” is intended to cover, for example, a computer system has circuitry that performs this function during operation, even if the computer system in question is not currently being used (e.g., a power supply is not connected to it). Thus, an entity described or recited as “configured to” perform some task

refers to something physical, such as a device, circuit, memory storing program instructions executable to implement the task, etc. This phrase is not used herein to refer to something intangible. Thus, the “configured to” construct is not used herein to refer to a software entity such as an application programming interface (API).

(12) The term “configured to” is not intended to mean “configurable to.” An unprogrammed FPGA, for example, would not be considered to be “configured to” perform some specific function, although it may be “configurable to” perform that function and may be “configured to” perform the function after programming.

(13) Reciting in the appended claims that a structure is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) for that claim element. Accordingly, none of the claims in this application as filed are intended to be interpreted as having means-plus-function elements. Should Applicant wish to invoke Section 112(f) during prosecution, it will recite claim elements using the “means for” [performing a function] construct.

(14) As used herein, the terms “first,” “second,” etc. are used as labels for nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.) unless specifically stated. For example, references to “first” and “second” service providers would not imply an ordering between the two unless otherwise stated.

(15) As used herein, the term “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect a determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. As used herein, the phrase “based on” is thus synonymous with the phrase “based at least in part on.”

DETAILED DESCRIPTION

(16) The description that follows includes exemplary systems, methods, techniques, instruction sequences, and/or computer program products that embody techniques of the present disclosure. However, it is understood that the described embodiments may be practiced without these specific details. For instance, although some examples refer to accessing tenant services, other types of service providers are contemplated, such as of Software-as-a-Service (SaaS), among others.

(17) In a multi-tenant software architecture described herein, each tenant can share the applications, data, and/or infrastructure. A multi-tenant software provider can use a multi-tenant software platform to provision applications and/or resources to multiple entities. The multi-tenant software platform can facilitate addition of new tenants and on-board data and/or services provided by these new tenants. The multi-tenant software platform can use identity services to provide access to these services, such as from an entity associated with one tenant to a service provided by another tenant. The multi-tenant software platform can also implement rules and policies for data access by various entities across the tenants (e.g., access by a first service of a first tenant to access a second service of a second tenant to perform a transaction).

(18) The multi-tenant software platform (also referred to as a multi-tenant platform) can include multiple tenants that have been previously on-boarded. The multi-tenant platform can facilitate access, using a unified identity associated with a user, to the entities, policies, and services of these tenants. The tenants can be hosted and managed by service providers. Access to the services can be determined based on privileges of the unified identity, as well as on policies of each of the tenants. The services can include various merchant services, such as in-store check-out (at a certain tenant) via a user device (associated with the unified identity), access to an online store of the tenant, order-ahead at a certain store of the tenant, a cash-in process (e.g., at an Automated Teller Machine (ATM)) at a certain tenant, a cash-out process (e.g., at an ATM), self-checkout for fuel at pay-at-

the-pump stations of the tenant, securely storing funding instruments for use in transactions (also referred to as vaulting funding instruments), among others. The services can include SaaS and Platform as a Service (PaaS) services and/or other cloud services that are accessible by the user device. For example, the solution provider can provide access, to the user device, to various software applications, or deliver such software to the user device.

(19) The multi-tenant platform can thus utilize unified identity services to provide access to selected services and/or data of the tenants based on policies associated with each of the tenants. The multi-tenant platform can use separate databases to store data to achieve isolation, such as when new tenants are on-boarded onto the existing platform, and providing logical and/or physical data isolation. Shared data access can be made possible by on-boarding the tenants and selectively cross-exposing services. For example, for a multi-tenant platform managed by PAYPAL, certain capabilities of the multi-tenant platform such as Risk-as-a-Service (RaaS) can be provided to certain users of on-boarded tenants such as XOOM. Similarly, certain capabilities of XOOM such as Fund Sending can be provided to certain users of PAYPAL and capabilities of BRAINTREE such as Vaulting can be provided to certain users of PAYPAL. Thus, in some embodiments, the multi-tenant platform can include a core tenant (such as PAYPAL) that offers core services and infrastructure (including Identity-as-a-Service (IaaS) functionality) and any additional data access to selected tenants and/or users accessing the multi-tenant platform from that tenant. The techniques disclosed herein, however, are not limited to providing access from a core tenant to another tenant. In various embodiments, any service of any tenant can be accessed by a service of another tenant subject to rules and/or policies as discussed herein. The level of access to the core services and/or infrastructure (or access to a tenant's services by services of other tenants) can be determined by rules and/or policies of each tenant and/or of the multi-tenant platform.

(20) The multi-tenant platform can onboard new tenants as needed by using a one or more hierarchical data structures. In some embodiments, the multi-tenant platform can expose appropriate user experiences for a user application based on the user request. The multi-tenant platform can implement methods for providing IaaS services to various entities and/or tenants. The multi-tenant platform can provide the IaaS and other services based on the hierarchical data structures. As discussed below, the multi-tenant platform can process user requests (e.g., from users of the core tenant or from users of an on-boarded tenant) and determine how to access a certain service.

(21) In some embodiments, the multi-tenant platform can receive a request, at a first service provider, to provide a service for a user. The multi-tenant platform can access a representation of a second service provider in a first hierarchical data structure managed by the first service provider. In some embodiments, the multi-tenant platform is the first service provider that can provide certain services to entities that are referenced via identifiers determined using IaaS. The multi-tenant platform can determine that user data required for the service is managed by the second service provider that manages user identity of the user. The multi-tenant platform can determine that the representation is linked with a full identity reference for the second service provider in a second hierarchical data structure managed by the second service provider. The multi-tenant platform can access the user data at the second hierarchical data structure using the full identity reference. The multi-tenant platform can access the service via the lightweight identity reference and using the user data at the first service provider. The following description and associated Figures illustrate various embodiments directed to the ideas listed above.

(22) FIG. 1 is a system diagram illustrating embodiments of a multi-tenant software architecture system that can be accessed by communicating with a user device. In an overview of the system diagram **100**, a user device **104** can communicate with multi-tenant platform **102**. The multi-tenant platform **102** can provide access to multiple service providers **108**, **110**, and **112**. The multi-tenant platform **102** includes core services at elements **132-148**, as well as additional services **130(1)-130(5)**. The multi-tenant platform **102** can provide services to the service providers **108-**

112, such as IaaS services that may be required for accessing other services. In various embodiments, the multi-tenant platform **102** can facilitate accesses by a service **116**, **120**, **124** of a service provider **108**, **110**, **112** to a service **116**, **120**, **124** of another service provider **108**, **110**, **112** (e.g., facilitating access by service **116** of service provider **108** to service **124** of service provider **112**). Thus, the multi-tenant platform **102** can manage a plurality of tenants, each one of which can be associated with one or more services, which are then exposed for access. In various embodiments, multi-tenant platform **102** is implemented using a computer system (e.g., a laptop computer, a desktop computer, a server) or a plurality of computer systems (e.g., a cloud of servers).

(23) The multi-tenant platform **102** can store user information for users at the accounts **142**. In some embodiments, the accounts **142** includes information for users (also referred to as “core users”) of a core tenant. The core tenant can be one of the service providers **108-112**, or a service provider that is fully integrated into the multi-tenant platform **102**. The multi-tenant platform **102** can facilitate provision of various core services to the core users. The core services can include identity services **132**, risk services **138**, compliance services **140**, and payment services **148**. In one embodiment, the multi-tenant platform **102** provides IaaS services at the identity services **132** element. The identity services **132** can generate and maintain a core hierarchical data structure for managing its core entities. The core services can also include access to policy configuration **136**, as well as access to accounts **142**, merchants **144**, and consumers **146**. The accounts **142** are associated with the identity services **132**.

(24) The service provider **108** can include entities **114**, policies **123**, and services **116**. Similarly, the service providers **110** and **112** include certain elements of entities **118** and **122**, policies **125** and **127**, and services **120** and **124**, respectively. In some embodiments, a certain service provider can be fully integrated onto the multi-tenant platform **102**, a process which can include generating the core hierarchical data structure for the core tenant. In this example, the identity services **132** can manage the core users, meaning that the service provider **108** does not have its own identity service for its own users. Alternatively, the service providers **108**, **110**, and **112** can be on-boarded onto the multi-tenant platform **102** and manage their own identities. Any access from a service provider **108**, **110**, **112** for access of a service other than its own services **116**, **120**, **124**; such as access of a service **130(1)** or access to a service **116**, **120**, **124** of another service provider **108**, **110**, **112**; uses a unified identity for access of services and/or data across the multi-tenant platform **102**, in various embodiments.

(25) The multi-tenant platform **102** can onboard the service providers **108**, **110**, **112**. As discussed below, the onboarding process can include accessing identity services of the services providers **108**, **110**, **112** to determine at least a portion of the entities that are being managed by the respective identity service. If the entities are not being migrated (e.g., by being fully integrated) to the identity services **132**, then the identity services **132** can generate representations (also referred to herein as models) of the entities being on-boarded in the core hierarchical data structure. In various embodiments, these representations are then used by the IaaS service to determine and use the unified identity (for a certain user) across the multi-tenant platform **102**. The representations can include lightweight elements and/or linked elements. The identity services **132** can manage and access the policy configuration **136**, such as to enforce access to certain customer and/or merchant domains. The policy configuration **136** can be accessed via the hierarchical data structures (discussed below) for each entity. The process of generating and using user representations is explained in more detail below.

(26) The multi-tenant platform **102** can communicate, via the multi-tenant API **126**, with the service providers **108**, **110**, **112** and/or the user device **104**. The multi-tenant platform **102** can provide, based on a certain unified identity, information related to a tenant, associated services, and/or associated user experiences, to a requesting entity via the multi-tenant API **126**.

Furthermore, the multi-tenant platform **102** can facilitate communication between various service

providers **108**, **110**, **112**, such as by providing IaaS services and use of unified identity. In various embodiments, multi-tenant API **126** includes one or more API gateways **126(1)**, **126(2)**, etc. (not shown) executable to interface with respective service providers **108**, **110**, **112** by, for example, handling protocol translations between the multi-tenant API **126** and the various service providers **108**, **110**, **112**. In some embodiments, for example, multi-tenant API **126** includes a first API gateway **126(1)** executable to translate protocols between service provider **108** and multi-tenant API **126** and a second API gateway **126(2)** executable to translate protocols between service provider **110** and multi-tenant API **126**. In various embodiments, such API gateways **126(n)** are implemented as components of API gateway **126** but in other embodiments may be implemented as separate instances of API gateway **126**.

(27) For example, a new tenant, such as the service provider **112**, can be on-boarded by the multi-tenant platform **102**. Information on the new tenant can be stored by one or more of the core services **132-148**. In some embodiments, one or more of services of the new tenant, such as the service **124**, can be exposed for access to other users of the multi-tenant platform **102**. In one embodiment, the multi-tenant platform **102** can also generate a representation **130(5)** of the service **124** at the multi-tenant platform **102**. Thus, any access from other tenants, such as from the user device **104** or from the service provider **110**, to the service **124** is performed via a unified identity (e.g., by using IaaS) at the service representation **130(5)**. As discussed herein, such accesses include the use of a universal access token (UAT) in various embodiments.

(28) The user device **104** can be any type of a computing device that can communicate user requests to the multi-tenant platform **102**. The user device **104** can be implemented as a kiosk, a point-of-sale (PoS) terminal, a mobile device, among others. The user device **104** includes a user interface (UI) **106** through which the user can interact with displayed user experiences, such as to access certain services provided via the multi-tenant platform **102**. The user device **104** can generate and communicate a user request for a certain service at the multi-tenant platform **102**. The user device **104** can interface with the multi-tenant API **126**, which can interface with various other components of the multi-tenant platform **102**.

(29) In some embodiments, one of the tenants, such as the core service provider **108**, can implement one or more payment system functions. While the embodiments described in this paragraph relate to embodiments in which one of the tenants is a core service provider that is involved in a transaction between respective services of two tenants, in various other embodiments neither tenant in the transaction is a core service provider **108**. In these embodiments, the core service provider (that is integrated into the multi-tenant platform **102**) can process a payment and order fulfillment for services provided via user experiences accessed at the UI **106**. The core service provider **108** can perform risk analysis (e.g., via the risk core service **138**) to determine whether or not to perform a certain service and/or process a payment for that service. In some embodiments, the core service provider **108** can perform risk and/or verification services for the new service provider **112** (i.e., a new tenant) being on-boarded by the multi-tenant platform **102**. The core service provider **108** can process payments from the user account that is associated with the user device **104**. The core service provider **108** can provide financial services, such as a fund transfer (e.g., a transfer of a certain monetary amount), to the user. The core service provider can include payment accounts, each of which can be associated with a buyer or a seller. Once on-boarded as a core tenant, the accounts of the core service provider can be stored and accessed directly at the multi-tenant platform at the accounts **142** element. For example, a buyer (e.g., a user of the user device **104**) can be associated with one payment account, and the seller (e.g., one of the tenants) can be associated with another payment account at the payment system. Upon successfully performing the risk analysis on the requested service (e.g., a requested transaction at the tenant), the core service provider can then perform (e.g., via the payment core service **148**) a fund transfer from the buyer's payment account to the seller's payment account.

(30) In various embodiments, two tenants (e.g., two of service providers **108**, **110**, and **112**) can

engage in transactions between their respective services as discussed in further detail herein in reference to FIGS. 5, 6A, 6B, and 7. In such embodiments, multi-tenant platform **102** receives (via multi-tenant API **126**) a request from a first service (e.g., one of services **116**, **120**, **124**) of a first tenant (e.g., one of service providers **108**, **110**, **112**) of the multi-tenant platform to access a second service (e.g., another of services **116**, **120**, **124**) of a second tenant (e.g., another of services providers **108**, **110**, **112**) to perform a transaction. In some embodiments, for example, the first service is a shopping cart service with which an end user identifies products for purchase from a merchant and the second service is a payment processing service that stores financial information relating to the end user and the merchant and is executable to transfer money from the end user to the merchant to pay for purchased products. As discussed herein, in various embodiments the first tenant and the second tenant have different systems by which the end user and merchant are identified and/or authenticated. In various embodiments, the request includes a first access token that is usable to authenticate the transaction with the first tenant, a universal access token is generated using the first access token, and a second access token useable to authenticate the transaction with the second tenant is generated using the universal access token, and the second access token is used to communicate with the second service to perform the transaction. Multi-tenant platform **102**, therefore, is able to facilitate the transaction even though the first tenant and second tenant (and their respective services) have different identification and/or authentication systems that, for example, use different formats that require different or additional information.

(31) The payment system can be implemented by PAYPAL or another online payment system that allows users to send, accept, and request fund transfers. In some embodiments, the user experience can also provide access to certain services. Thus, in addition, or instead of, a payment service, the user experience can include other functionality that is unique for the certain tenant, such as selection of items for order, access to certain SaaS functionality, the ability to vault various funding instruments with services implemented by BRAINTREE (a separate tenant), among others. Thus, the solution provider can provide funds in/funds out services, as determined based on the location data, from the payment system to the users of the user devices.

(32) For simplicity, FIG. 1 only shows a single user device **104**. However, as discussed herein, the multi-tenant platform **102** interfaces with multiple user devices, and thus provides access to services to many different users. Similarly, the multi-tenant platform **102** can onboard multiple tenants in addition to the ones shown. The multi-tenant platform **102** can also have multiple core tenants that are incorporated as part of the core services **132-148**. As discussed herein, multi-tenant platform **102** facilitates an end consumer's ability to access checkout services via Merchants interfacing with different Service Providers (e.g., a first service provider for checkout and a second service provider for payment processing), in various embodiments.

(33) FIG. 2 is a system diagram illustrating embodiments of the multi-tenant software architecture of FIG. 1 including unified identity services. FIG. 2 shows embodiments of how the identity services are implemented by each of the service providers **108-112**, as well as by the multi-tenant platform **102**. As shown, the service provider **108** can include a third-party identity provider (3P-IDP) **204** that manages entities of merchant **206(1)** and **206(2)**, and entities of consumer **208(1)** and **208(2)**. Similarly, the service provider **110** can include a 3P-IDP **204** that manages entities of consumer **214(1)** and **214(2)**. The service provider **112** can include a 3P-IDP **222** that manages entities of merchant **224(1)** and **224(2)**. The multi-tenant platform **102** includes the identity service **132**, the accounts **142** element, the risk **138** element, as well as services **130(1)** and **130(2)**. In some embodiments, the 3P-IDPs **204**, **212**, and **222** are representations (in the hierarchical data structure **244**) of the actual IDPs provided by the service providers **108**, **110**, and **112**.

(34) In the embodiments shown in FIG. 2, the service provider **108** can be on-boarded by the multi-tenant platform **102**, but not fully integrated. Thus, a link **211** is shown that indicates the on-boarded but not fully integrated status of the service provider **108**. Alternatively, service providers can be fully integrated with the multi-tenant platform **102** by being made core to the multi-tenant

platform **102**. In various embodiments, identity service **132** includes an access tier **242** and the hierarchical data structure **244** that is managed by a multi-tenant identity provider (MT-IDP) **246**. The multi-tenant platform **102** also includes services **138**, **142**, and **130(1)-130(2)**.

(35) The access tier **242** is a part of the identity service **132** that customizes data associated with the service provider when being on-boarded with the multi-tenant platform **102**. For example, the access tier **242** includes information about the products and/or services being offered by the service provider **108**, as well as service endpoints for accessing services offered by the service provider **108**. The service endpoints can be referenced by a corresponding node in the hierarchical data structure **244** for the service provider **108**. Various embodiments of how the hierarchical data structure **244** is generated and accessed are discussed below with reference to FIGS. **3** and **4**. In various embodiments, MT-IDP **246** can generate and manage representations of the entities of the service provider **108** being on-boarded (e.g., entities **206(1)**, **206(2)**, **208(1)**, and **208(2)**). Similarly, MT-IDP **246** can generate and manage representations of entities for tenants of service providers **110** and **112**.

(36) As discussed, since each of the service providers **108**, **110**, **112** includes its own 3P-IDP **204**, **212**, and **222** respectively, each of these tenants continues to manage the identities of their own respective entities. For example, the service provider **110** continues to manage the entities **214(1)-214(2)**, including managing identity information, contact data, and any characteristics of the consumers **214(1)** and **214(2)**. The service provider **110** can update any changes to the entities **214(1)** and **214(2)** to the multi-tenant platform **102**. These 3P-IDP **212** originated updates are used to keep the data of their respective representations up-to-date. Similarly, any services (such as core services) performed via the representations of the hierarchical data structure **244** are also propagated back to the 3P-IDP **212**. As discussed further in reference to FIGS. **5**, **6A**, and **6B**, in various embodiments, respective 3P-IDPs **204**, **212**, **222** manage access tokens and internal representation of access (also referred to herein as “security contexts”) associated with their respective service providers (e.g., first access token **530** discussed with reference to FIG. **5** is exchangeable for security context by 3P-IDPs **204** **204**). In various embodiments, MT-IDP **246** is executable, using hierarchical data structure **244** and respective security contexts from 3P-IDPs **204**, **212**, **222**, to generate a universal access token using the first token **530**. Using the universal access token (e.g., universal access token **532** discussed with reference to FIG. **5**), the 3P-IDP of the second service provider (e.g., 3P-IDP **222** or service provider **112**) generates an access token for the second service provider that is useable to authenticate the transaction with the second service (e.g., second access token **534** discussed with reference to FIG. **5**). In such embodiments, multi-tenant API **126** (shown in FIG. **1**) is executable to use the second access token to communicate with the second service (e.g., another of services providers **108**, **110**, **112**) to perform the transaction.

(37) FIG. **3** is a diagram illustrating a hierarchical data structure **244** used by multi-tenant platform **102** to provide unified identity services. FIG. **3** illustrates a hierarchical data structure **244** that is managed by the MT-IDP **246**. In FIG. **3**, there are no additional IDPs, and the entities **301**, **302**, **304**, **306**, **310**, **312**, **216**, **318**, **324**, **326**, and **328** are fully integrated into the multi-tenant platform **102**. In FIG. **3**, a first layer of hierarchy includes the service provider **301**, which can correspond to the fully integrated core service provider **108** (e.g., PAYPAL), and an additional service provider **308**, which represents a separate tenant (e.g., service provider **110** or **112**) of multi-tenant platform **102**. The organization and linking of the hierarchical data structure **244** of FIG. **3** can be referred to as a dependency graph. Although not shown in FIG. **3** (or FIG. **4**), each of the merchants can include, as child nodes, a plurality of users such as consumers. Other nodes in FIGS. **3** and/or **4** can similarly have additional children nodes, as desired. As seen in FIGS. **3** and **4**, the hierarchical data structure **244** can be used by the multi-tenant platform **102** to model various organizations including small merchant, large merchant, bring-your-own merchant, bring-your-own-customer, marketplaces, among others.

(38) The next layers of hierarchy include the service providers **304** and **306**. The service providers **304** and **306** can be on-boarded but not fully integrated. The service providers **304** and **306** can inherit policies and configuration from the service provider **301**. The hierarchical data structure shows that the merchants **316** and **318** are child nodes of the node of the service provider **306**, and thus are acted upon by the service provider **306**. The service provider **306** (as inherited from the service provider **301** and modified by access permissions associated with the service provider **306**) defines the permissions of the merchants, including how the merchants **316** and **318** can communicate with other entities of the hierarchical data structure including consumers of the merchant **316**.

(39) The hierarchical data structure shows that the consumers **310** and **312** are child nodes of the node of the merchant **302**, and thus are acted upon by the merchant **302**. The merchant **302** (as inherited from the service provider **301** and modified by access permissions associated with the merchant **302**) defines the permissions of the consumers, including how the consumers **310** and **312** can communicate with other entities of the hierarchical data structure **244**. For example, the merchant **302** can define that the consumer **310** can view its own profile, access certain core services (such as services **130(1)** and **130(2)**). The service provider **306** can be associated with small business merchants, and thus merchants **316** and **318** can each represent a small business merchant in the embodiment shown in FIG. 3.

(40) The service provider **306** includes child nodes of merchants **316** and **318**. The hierarchical data structure shows that the merchants **316** and **318** are linked to (but are not child nodes of) the service provider **304**. Thus, the merchants **316** and **318** can be acted upon by the service provider **306**, but can also be accessed (such as with read but not write privileges) by the service provider **304**. An example of this is where the merchants **316** and **318** are two merchants that are operating as a part of the service provider **306** (which can implement a certain branch of PAYPAL). In this example, the service provider **304** can be a marketplace tenant such as EBAY that includes one entity (merchant **302**), as well as links to two merchants **316** and **318**. The service provider **306** (as inherited from the service provider **301** and modified by access permissions associated with the service provider **306**) defines the permissions of the merchants, including how the merchants **316** and **318** can communicate with other entities of the hierarchical data structure including consumers of the merchant **316**. It is noted that these links are used to link with nodes representing merchants **316** and **318**, instead of using representations (as discussed below) since the same MT-IDP **246** manages all of the nodes of **316**, **318**, and **306**. FIG. 3 also illustrates administrators **330** and **332** that have access to the merchant **302** and to the service provider **306**, respectively. The administrator **330** and **332** can have various access permissions (such as ability to read configuration data).

(41) FIG. 4 is another diagram illustrating hierarchical data structures **244** used by multi-tenant platform **102** to provide unified identity services. FIG. 4 illustrates several hierarchical data structures **402-406**. The first hierarchical data structure **402** is managed by the MT-IDP **246**. The second hierarchical data structure **404** is managed by the 3P-IDP **212**, and the third hierarchical data structure **406** is managed by the 3P-IDP **232**. FIG. 4 illustrates the relationships between the entities of the hierarchical data structures **402-406**. The organization and linking of the hierarchical data structures of FIG. 4 can be referred to as a dependency graph. It is noted that the organization and type of the hierarchical data structures **404** and/or **406** is shown for illustrative purposes only, and that one or more of the 3P-IDP **212** and **222** can implement different data structures, as desired. In this case, the Identity-as-a-Service (IaaS) services can further map different schema used by the 3P-IDP **212** and/or **222** to the hierarchical data structure **402**. It is also noted that the hierarchical data structure managed by the MT-IDP **246** is also shown for illustrative purposes only, and other implementations are contemplated. In some embodiments, the 3P-IDPs **212** and **222** are representations (in the hierarchical data structure **244**) of the actual IDPs provided by the service providers **110** and **112**.

(42) In FIG. 4, the entities **412(1)**, **412(2)**, **412(3)**, **414(1)**, **414(2)**, **413(1)**, **416**, **418**, **419**, **420(1)**, and **420(2)** are fully integrated into the multi-tenant platform **102**, and are managed by the MT-IDP **246**. The entities **214(1)** and **214(2)** are not fully integrated into the multi-tenant platform **102**, and are instead on-boarded onto the multi-tenant platform **102**. The entities **214(1)** and **214(2)** are managed by the 3P-IDP **212**. Similarly, the entities **224(1)**, **224(2)**, and **224(3)** are on-boarded but not fully integrated into the multi-tenant platform **102**. The entities **224(1)**, **224(2)**, **224(3)** are managed by the 3P-IDP **232**. In various embodiments, service provider **412(1)** of FIG. 4 represents service provider **110** of FIG. 1 in hierarchical data structure **244**, service provider **412(2)** of FIG. 4 represents service provider **108** of FIG. 1 in hierarchical data structure **244**, and service provider **412(3)** of FIG. 4 represents service provider **112** of FIG. 1 in hierarchical data structure **244**.

(43) In various embodiments, multi-tenant platform **102** generates representations of merchant **418** and merchant **419** under service provider **412(2)** (that are managed by MT-IDP **246**). In various embodiments, merchant **418** represents a particular identity under service provider **412(2)** (e.g., corresponding to a first particular account at service provider **108** that, for example, is associated with a first ecommerce storefront) and merchant **419** represents a separate identity under service provider **412(2)** (e.g., corresponding to a second account at service provider **108** that, for example, is associated with a second ecommerce storefront). In various embodiments, an additional service provider **413(1)** is also represented under service provider **412(2)**. In various embodiments, additional service provider **413(1)** operates as a subpart of service provider **412(2)** (e.g., as an ecommerce service provider operating within another ecommerce service provider such as ETSY operating as a subpart of PAYPAL). In such embodiments, additional service provider **413(1)** has child nodes (not shown) representing merchants and/or consumers.

(44) During the process of onboarding the service provider **110**, the multi-tenant platform **102** can generate representations for the consumers **214(1)** and **214(2)** (that are managed by that tenant's 3P-IDP **212**). The multi-tenant platform **102** can generate these representations by generating linked consumers **414(1)** and **414(2)**. A linked consumer is an entity that represents a corresponding entity in another hierarchical data structure, and that contains a limited amount of the data for that corresponding entity. The linked consumers **414(1)** and **414(2)** link to the consumers **214(1)** and **214(2)**, respectively.

(45) Similarly, during the process of onboarding the service provider **112**, the multi-tenant platform **102** can generate representations for the merchants **224(1)** and **224(2)** that are managed by that tenant's 3P-IDP **222**. The multi-tenant platform **102** can generate these representations by generating lightweight merchants **420(1)** and **420(2)**. A lightweight merchant is an entity that represents a corresponding entity in another hierarchical data structure, and that contains some of the data for that corresponding entity. The lightweight merchants **420(1)** and **420(2)** link to the merchant **224(2)** and lightweight merchant **224(3)**, respectively. The types of representations can include a linked entities and lightweight entities. In some embodiments, a linked entity (e.g., a linked consumer or a linked merchant) is different from a lightweight entity (e.g., a lightweight merchant or a lightweight consumer) by is that the lightweight entity includes an additional amount of information for its corresponding entity, and the type can be chosen during onboarding.

(46) The hierarchical data structure **402** indicates that the linked consumers **414(1)** and **414(2)** are child nodes of the node of the service provider **412(1)**, and thus are acted upon by the service provider **412(1)**. However, the service provider **412(1)** does not directly modify the linked to consumers **214(1)** and **214(2)**. Additionally, the hierarchical data structure **402** indicates that merchant **418** and merchant **419** are child nodes of the node of the service provider **412(2)**. Similarly, the hierarchical data structure **402** indicates that the lightweight merchants **420(1)** and **420(2)** are child nodes of the node of the service provider **412(3)**, and thus are acted upon by the service provider **412(3)**. However, the service provider **412(3)** does not directly modify the linked to merchants **224(1)** and **224(2)**. Additionally, merchant **418** is linked to lightweight merchant **420(1)** and merchant is linked to lightweight merchant **420(2)**, indicating that these linked entities

within hierarchical data structure **402** are associated with associated identities with service provider **108** and service provider **112**, respectively (e.g., merchant **418** and lightweight merchant **420(1)** are both controlled by the same person or group).

(47) An example of this is where the merchants **224(1)** and **224(2)** are operating as merchants of a service provider 3P-IDP **222** (it is noted that for simplicity, both the 3P-IDP **222** and its service provider are shown as one box in FIG. **4**, and similarly for the 3P-IDP **212**), which can be a marketplace tenant. Although the lightweight merchants **420(1)** and **420(2)** inherit the access permissions from the service provider **412(2)**, the 3P-IDP **222** provides the access permissions to the merchants **224(1)** and **224(2)**. In case of access permission conflict, the access permissions of the merchants **224(1)** and **224(2)** prevails.

(48) As discussed in further detail in reference to FIG. **5**, the hierarchical data structures **244** depicted in FIGS. **3** and **4** are useable to facilitate transactions between services (e.g., services **116** and **124**) of various service providers (e.g., service provider **108** and service provider **112**). For example, in one embodiment, a particular merchant is implemented under service **116** of service provider **108** (e.g., PAYPAL) and therefore has access credentials associated with service provider **108** (e.g., first token **530** shown in FIG. **5**). In such embodiments, this particular merchant is modeled under service provider **412(2)** as merchant **418**.

(49) In various embodiments, in order to complete a transaction between two different services from different service providers, the party or parties (e.g., end users such as merchants or customers) to the transaction are authenticated with both services and service providers. In various embodiments, however, the authentication systems used by these different services and service providers are different and require different information and/or information in different formats. For example, in an embodiment, service provider **108** uses a user name and a password received from the end users for authentication, and upon authenticating the end user, generates a first cryptographic access token having a first format (e.g., first token **530** shown in FIG. **5**) for the transaction. In this embodiment, service provider **110** may require different information to authenticate the end user (e.g., a different user name, a different password, and a second factor authentication) and/or the second cryptographic access token required to complete the transaction (e.g., a second token **534** shown in FIG. **5**) is in a different format and/or requires different information. Accordingly, in various embodiments, the hierarchical data structures **244** include multiple representations of the same end user(s) associated with different tenants and links between representations associated with the same end user.

(50) Using the hierarchical data structures **244** maintained by MT-IDP **246** and respective 3P-IDPs (e.g., 3P-IDP **204** and **222**), multi-tenant platform **102** is operable to receive the first cryptographic access token (e.g., first token **530** shown in FIG. **5**) and generate the second cryptographic access token (e.g., a second token **534** shown in FIG. **5**). As discussed herein, the hierarchical data structures **244** model the authentication systems of the various service providers (e.g., how an access token is usable to authenticate the transaction with the respective tenants, what information is used and in what format for the access tokens), and using these hierarchical data structures **244**, multi-tenant platform **102** is operable to generate the second cryptographic access token (e.g., by adding additional information to the information received in the first cryptographic access token, putting such information in the proper format for the second cryptographic access token).

(51) FIG. **5** is a diagram illustrating embodiments of operations for facilitating a transaction between service providers by the multi-tenant platform **102** using unified identity services. The operations of FIG. **5** are described with reference to the systems and components described in FIGS. **1-4** (for illustration purposes and not as a limitation). The example operations can be carried out by the multi-tenant platform **102** in various embodiments.

(52) User device(s) **104** interface with merchant **501** and **503** at **500** and **502**. In the embodiments shown in FIG. **5**, merchant **501** is implemented via service provider **108** and merchant **503** is implemented via service provider **112**. In these embodiments, such interfacing includes one or

more user devices **104** accessing the respective service providers **108**, **110** via the Internet (e.g., by accessing a website associated with the service providers **108**, **112**) and selecting merchant **501** and merchant **503** to facilitate a transaction. In various embodiments, different user devices **104** are used by different users (e.g., a first user device **104** associated with a customer interfacing with service provider **108** and a second user device **104** associated with a merchant interfacing with service provider **112**). In other embodiments, the same user device **104** interfaces with both service provider **108** and service provider **112** at **500** and **502**. In various embodiments, for example, at **500** a user device **104** interfaces with a website associated with merchant **501** on service provider **108** (e.g., an e-commerce website) to choose products for purchase at **500** and interfaces with merchant **503** on service provider **112** (e.g., a payment processing service) to select a payment method for the transaction (e.g., to transfer funds from the customer to the merchant to pay for the products) at **502**.

(53) While FIG. 5 includes merchants **501** and **503**, the disclosed techniques are not limited to transactions involving merchants, however. In various embodiments, the merchant **501** can be any user of service provider **108** and merchant **503** can be any user of service provider **112**. In various embodiments, merchant **501** and merchant **503** are linked in hierarchical data structure **244** (e.g., merchant **501** and merchant **503** are associated with the same entity). In other embodiments, merchant **501** and merchant **503** are not associated with the same entity. As discussed herein, using the techniques disclosed herein, merchant **501** is able to leverage services **116** and **124** offered by respective service providers **108** and **112** during the transaction. In various embodiments, merchant **501** receives first token **530** discussed below from service provider **108**.

(54) At **504**, on behalf of merchant **501**, service provider **108** interfaces with multi-tenant platform **102** via multi-tenant API **126** to request to use first service **116** and to access a second service of a different service provider (e.g., service **124** of service provider **112**) to perform a transaction. In various embodiments, this interfacing includes first token **530** discussed below. Multi-tenant API **126** is illustrated three times in FIG. 5 connected by dashed lines: a first interfacing with service provider **108**, service **116**, and 3P-IDP **204** at **504**, **510**, and **506**, respectively; a second interfacing with service **116** and service **124**; and a third interfacing with service provider **112**, service **124**, and 3P-IDP **222**. In various embodiments, these interfaces are implemented with multiple instances of multi-tenant API **126**. In such embodiments, a first instance of multi-tenant API **126(1)** is useable to facilitate communications between multi-tenant platform **102** and first service **116** and a second instance of multi-tenant API **126(2)** is useable to facilitate communications between multi-tenant platform **102** and second service **124** as shown in FIG. 5. In various other embodiments, multi-tenant API **126** is implemented using separate APIs (e.g., a first multi-tenant API **126** for first service provider **108**, a second multi-tenant API **126** (not shown) for second service provider **112**).

(55) At **506**, multi-tenant API **126(1)** interfaces with 3P-IDP **204** (i.e., the 3P-IDP **204** of service provider **108**) to request exchange of first token **530** to an internal token (also referred to herein as a “security context”) that is useable by multi-tenant API **126(1)** make calls to the services of service provider **108** (e.g., service **116**). In response to the request to exchange first token **530** with the security context for service **116**, 3P-IDP **204** communicates with MT-IDP **246** at **508** to determine service provider **108** and service provider **112** are provisioned within hierarchical data structure **244** and that merchant **501** is provisioned under service provider **108** (e.g., under service provider **412(2)**) and that merchant **503** is provisioned under service provider **112** (e.g., under service provider **412(3)**). Further, a determination is also made whether a representation of merchant **503** is provisioned under the hierarchical data structure maintained by 3P-IDP **222**. If such representations are not provisioned, 3P-IDP **204**, MT-IDP **246**, and/or 3P-IDP **222** work to provision the missing representations. In the embodiment shown in FIG. 5, M1' under service provider **412(2)** corresponds to merchant **501** and M1' under service provider **412(3)** and 3P-IDP **222** correspond to merchant **503**.

(56) As discussed herein in reference to FIGS. 3 and 4, 3P-IDP **204**, 3P-IDP **222**, and MT-IDP **246**

generate hierarchical data structure(s) **244** associated with service providers **108**, **112** and/or add nodes to existing hierarchical data structure(s) **244** associated with the transaction (e.g., by adding nodes associated with end users associated with the transaction). In various embodiments, the hierarchical data structure(s) **244** are provisioned in prior to the request for the transaction is received. In other embodiments, hierarchical data structure(s) **244** are generated during the transaction. After determining that the hierarchical data structure **244** is properly provisioned, 3P-IDP returns the security context for service **116** to multi-tenant API **126(1)**. Using the returned security context, multi-tenant API **126(1)** calls service **116** at **510**.

(57) If merchant **503** is not provisioned under 3P-IDP **222**, MT-IDP **246** sends a request for 3P-IDP **222** to provision merchant **503** under 3P-IDP **222** at **516**. In response, 3P-IDP **222** provisions merchant **503** in hierarchical data structure **244** and returns to MT-IDP **246** an external-identifier correlating to the merchant **503** under 3P-IDP **222**. In various instances, the actions performed at **516** only occur when the merchant **503** is not provisioned under 3P-IDP **222** and thus not performed with the same merchant **503** is invoked in subsequent transactions.

(58) At **512**, service **116** requests a universal access token **532** for the transaction from 3P-IDP **204**. At **514**, 3P-IDP **204** sends a request to MT-IDP **246** for a universal access token **532**. In response to **514**, MT-IDP **246** returns to 3P-IDP **204** a universal access token **532** as discussed below. In turn, 3P-IDP **204** sends the generated universal access token **532** to service **116**.

(59) In turn, using universal access token **532**, service **116** interfaces with multi-tenant API **126(2)** at **518**. Multi-tenant API **126(2)** sends universal access token **532** to 3P-IDP **222** at **520**. 3P-IDP **222** verifies the universal access token **532** and uses it to generate a second token **534**. In various embodiments, such verification and generation includes (a) accessing the hierarchical data structure(s) **244** associated with the transaction (e.g., associated with end users involved in the transaction) and (b) generating second token **534** to include information needed by service **124** to authenticate the transaction (e.g., what information should be provided in by second token **534**, in what format second token **534** should be, how to interpret the various portions of by second token **534**, cryptographic keys associated with second service provider **112** to decrypt and/or authenticate by second token **534**, etc.).

(60) At **522**, multi-tenant API **126** calls service **124** (e.g., by calling an API for service **124**) using the second access token **534** to complete the transaction. Once the transaction is completed, service **124** sends a transaction complete message to multi-tenant API **126(2)** which in turn sends a transaction complete message to service **116**. Service **116** then indicates that the transaction was completed to merchant **501** via multi-tenant API **126(1)**.

(61) In various embodiments, first token **530** and second token **534** are external access tokens that contain security credentials for the transaction and identifies the end user(s) and end user's access privileges. In various embodiments, first token **530** is generated by a logon service of service provider **108** when an end user logs on to service provider **108** (e.g., by providing credentials) and the end user is authenticated (e.g., by comparing the provided credentials against an authentication database). As discussed herein, first token **530** is a cryptographic access token that is usable to authenticate the transaction with service provider **108** and is in a first format used by the first service **116**, in various embodiments. Similarly, second token **534** is a cryptographic access token that is useable by service **124** to authenticate the transaction with service provider **112**. In various embodiments, second token **534** is in a second format used by second service **124**. In various embodiments, first token **530**, second access token **534**, and universal access token **532** are in different formats. In contrast to access tokens such as first token **530** and second token **534**, in various embodiments, respective 3P-IDPs **204**, **212**, **222** use respective access tokens (e.g., 3P-IDP **204** uses first token **530**) to generate a security context that is passed within multi-tenant platform **102** and are used to generate universal access tokens **532** as discussed herein.

(62) In various embodiments, universal access token (UAT) **532** is a compact and self-contained object useable to securely transmit information through various components of multi-tenant

platform **102** securely and to grant access to resources (e.g., resources managed by services providers **108** and **112**). In various embodiments, UAT **532** is a JSON web token (JWT) that is digitally signed using one or more secrets or one or more public/private key pairs. In various embodiments, UAT **532** contains information from (or generated using) first token **530** (e.g., information extracted by 3P-IDP **204** and used to generate a security context that is in turn used to generate UAT **532**) that is useable to authenticate a transaction with service provider **108** in addition to information that, when included in a second token **534**, is useable to authenticate the transaction with service provider **112**. In various embodiments, UAT **532** is a JWT as defined by RFC 7519 published by the Internet Engineering Task Force (IETF) dated May 2015. As discussed herein, using UAT **532**, multi-tenant platform **102** is operable to exchange tokens (e.g., first token **530** and second token **534**) from one domain to another (e.g., from service provider **108** to service provider **112**).

(63) In various embodiments, UAT **532** includes a header that identifies the UAT **532** (e.g., by identifying the type of token as a JWT, by indicating the signing algorithm used), a payload, and a signature. In various embodiments, the signature is used to verify that UAT **532** was not altered during transmission. In embodiments in which UAT **532** is signed with a private key, the signature can also verify that identity of the sender of UAT **532**. The payload of UAT **532** contains the information used to authenticate the transaction with service provider **108** and service provider **112**. In various embodiments, the payload includes an issuer indication of the tenant that requested the transaction (e.g., service provider **108**), an audience indication of the tenant that is receiving the transaction (e.g., service provider **112**), and indicators associated with the transaction. In various embodiments, such transaction indicators of the end users include first tenant payload information including a first indication of a first end user within first service **116** (e.g., an indicator of merchant **501**) and second tenant payload information including a second indication of the first end user within the second service **124** (e.g., an indicator of merchant **503**). For example, this first indication and second indication identify an entity selling products to a customer of the transaction in both first service **116** (e.g., an indicator of merchant **501**, an indicator of the products sold) and second service **124** (e.g., an indicator of merchant **503**, an indicator of an account to receive resources for the same). In some of such embodiments, the payload also includes a third indication of a second end user within first service **116** and/or second service **124** to identity the customer of the transaction.

(64) In various embodiments, the format of the UAT **532** is independent of the formats of access tokens used by any of the tenants of multi-tenant platform **102**. Accordingly, if multi-tenant platform **102** received an additional request from the first service **116** to access a third service (e.g., service **124**) of a third tenant of the multi-tenant system (e.g., service provider **112**) to perform an additional transaction, the UAT **532** generated to facilitate the additional transaction would have the same format as the UAT **532** shown in FIG. 5. Because in various embodiments the format of the tokens (and the authentication method) used by the respective services providers is proprietary, exchanging the various tokens between domains using a UAT **532** allows the multi-tenant platform **102** to accommodate new tenants (and facilitate transactions between them) in a more scalable way. The resources used to on-board a new tenant and/or set up a new interface between services of different tenants are reduced because rather than adding conversion algorithms between the token used by the new tenant or new service and the various other tenants and services, the new tenant merely can set up a conversion algorithm between its token and UAT **532**. For example, if a multi-tenant platform **102** includes five tenants that have their own services, facilitating transactions between all five services would require ten conversion algorithms (e.g., first service to second service, first service to third service, second service to third service, etc.). In contrast, if the tokens used by the respective services are exchanged via UAT **532** as described herein, only five conversion algorithms are required (e.g., first service to UAT **532**, second service to UAT **532**).

(65) FIGS. 6A and 6B are flow diagrams illustrating embodiments of operations of a method **600**

using multi-tenant platform **102** to facilitate transactions between services of different tenants. The method **600** of FIGS. **6A** and **6B** is described with reference to the systems and components described in FIGS. **1-5** (for illustration purposes and not as a limitation). The example operations can be carried out by the multi-tenant platform **102**. In the embodiment shown in FIGS. **6A** and **6B**, the hierarchical data structure(s) **244** are generated (at arrows **604-618**) for use in the transaction after the request for the transaction has been received (at arrow **602**), but in other embodiments the hierarchical data structure(s) **244** and the information contained therein may be provisioned prior to the request to perform the transaction being received (at arrow **602**). In FIGS. **6A** and **6B**, multi-tenant API **126** is represented in two instances: **126(1)** and **126(2)** as discussed here. As discussed above, however, in other embodiments, multiple, separate multi-tenant APIs **126** are used. Further, method **600** is performed in a context in which the hierarchical data structure **244** has not been fully provisioned for the transaction, but in other instances the hierarchical data structure **244** has already been generated, and thus certain tasks may be omitted in embodiments. In still other embodiments, a UAT **532** for a first transaction has been retained and is reused for a second transaction, and thus certain tasks can be omitted.

(66) In the embodiment shown in FIGS. **6A** and **6B**, first service provider **108** is an e-commerce platform (e.g., used by customers to select items for purchase from merchants via an e-commerce service **116**) and second service provider **112** is a payments processing platform (e.g., used by customers to securely store funding instruments such as credit card accounts and use them for payments during the checkout). These techniques, however, are not limited to financial transactions and could be used to perform a transaction in which computer files (e.g., electronic medical records, media files) are securely transferred from a first storage service to a second storage service, for example.

(67) At arrow **602**, first service provider **108** calls the multi-tenant API **126(1)** with an access token (e.g., first token **530**) as part of a request to perform a transaction to purchase products. At arrow **604**, multi-tenant API **126(1)** communicates with first 3P-IDP **204** (i.e., 3P-IDP of service provider **108**) to begin generation of a hierarchical data structure **244** for service provider **108** and service provider **112**. At arrow **606**, first 3P-IDP **204** communicates with MT-IDP **246** to generate merchant representations (e.g., representations **224(1)** and **224(2)** shown in FIG. **4**) for one or more merchants in the transaction within hierarchical data structure **244** on behalf of both first service provider **108** and second service provider **112**. At arrows **608** and **610**, MT-IDP **246** communicates with second 3P-IDP **222** via multi-tenant API **126** to provision one or more light weight merchants (e.g., lightweight merchants **420(1)** and **420(2)** associated with representations **224(1)** and **224(2)**) as shown in FIG. **4**) and then returns at arrow **612**. At arrow **614**, multi-tenant API **126** communicates information (e.g., an external ID) representing the merchants (e.g., lightweight merchants **420(1)** and **420(2)**) to MT-IDP **246** to include in the UAT **532**. At arrow **616**, MT-IDP **246** communicates the generated hierarchical data structure **244** to first 3P-IDP **204**. At arrow **618**, 3P-IDP **204** returns the security context useable to access service **116** for the transaction to multi-tenant API **126(1)**.

(68) At arrow **620**, multi-tenant API **126(1)** call first service **116** using the security context. At arrow **622**, first service **116** sends a request to MT-IDP **246** to translate the access token from first service provider **108** (e.g., first token **530**) to a UAT **532**. At arrow **624**, after generating the UAT **532** using the hierarchical data structure **244**, MT-IDP **246** sends UAT **532** (including the payload information for the transaction) to first service **116**.

(69) At arrow **626**, first service **116** sends UAT **532** to multi-tenant API **126(2)** so an access token for second service provider **112** (e.g., second token **534**) can be generated. At arrow **628**, multi-tenant API **126(2)** sends UAT **532** to second 3P-IDP **222** (i.e., 3P-IDP of service provider **112**) to validate UAT **532** and to generate second token **534**. At arrow **630**, second 3P-IDP **222** communicates with multi-tenant API **126(2)** and indicates that the validation of UAT **532** was successful and second token **534** was successfully generated. At arrow **632**, multi-tenant API

126(2) uses second token **534** to communicate with second service **124** (e.g., via an API call to second service **124**) to perform the transaction (e.g., to transfer financial resources from a customer to a merchant to pay for purchased products). At arrow **634**, second service **124** returns to multi-tenant API **126(2)** (e.g., at an API gateway for second service **124**). At arrow **636**, multi-tenant API **126(2)** returns to first service **116**. At arrow **638**, first services **116** returns to multi-tenant API **126(1)** (e.g., at an API gateway for first service **116**). At arrow **640**, multi-tenant API **126(1)** returns to first service provider **108** completing the transaction operation.

(70) Referring now to FIG. 7, a flow diagram depicting a transaction method **700** is depicted. In the embodiment shown in FIG. 7, the various actions associated with method **700** are implemented by multi-tenant platform **102**. At block **702**, multi-tenant platform receives a request from a first service (e.g., service **116**) of a first tenant of the multi-tenant platform **102** (e.g., service provider **108**) to access a second service (e.g., service **124**) of a second tenant of multi-tenant platform **102** (e.g., service provider **112**) to perform a transaction. The request includes a first access token (e.g., first token **530**) that is usable to authenticate the transaction with the first tenant (e.g., service provider **108**) and is in a first format used by the first service (e.g., service **116**). At block **704**, multi-tenant platform **102** generates, using the first access token (e.g., first token **530**), a universal access token **532**. At block **706**, multi-tenant platform **102** generates, using universal access token **532**, a second access token (e.g., second token **534**). The second access token is useable to authenticate the transaction with the second tenant (e.g., service provider **112**) and is in a second format used by the second service (e.g., service **124**). At block **708**, multi-tenant platform **102** uses the second access token (e.g., second token **534**) to communicate with the second service (e.g., service **124**) to perform the transaction.

(71) It should be understood that FIGS. **1-7** and the operations described herein are examples meant to aid in understanding embodiments and should not be used to limit embodiments or limit scope of the claims. Embodiments may perform additional operations, fewer operations, operations in a different order, operations in parallel, and some operations differently. For example, one or more elements, steps, or processes described with reference to the flow diagrams of FIG. **5**, **6A**, **6B**, or **7** may be omitted, described in a different sequence, or combined as desired or appropriate.

(72) As will be appreciated by one skilled in the art, aspects of the present disclosure may be embodied as a system, method, or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, a software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(73) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible and/or non-transitory medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(74) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave.

Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device.

(75) Computer program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(76) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer program code may execute (e.g., as compiled into computer program instructions) entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(77) Aspects of the present disclosure are described with reference to flow diagram illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the present disclosure. It will be understood that each block of the flow diagram illustrations and/or block diagrams, and combinations of blocks in the flow diagram illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the computer program instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flow diagrams and/or block diagram block or blocks.

(78) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flow diagram and/or block diagram block or blocks.

(79) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flow diagrams and/or block diagram block or blocks.

(80) FIG. 8 is a block diagram of one embodiment of an electronic device **800** used in the communication systems of FIGS. 1-6. In some implementations, the electronic device **800** may be a laptop computer, a tablet computer, a mobile phone, a kiosk, a powerline communication device, a smart appliance (PDA), a server, and/or one or more other electronic systems. For example, a user device may be implemented using a mobile device, such as a mobile phone or a tablet computer. For example, a payment system may be implemented using one or more servers. The electronic device **800** can include a processor unit **802** (possibly including multiple processors, multiple cores, multiple nodes, and/or implementing multi-threading, etc.). The electronic device **800** can also include memory unit **806**. The memory unit **806** may be system memory (e.g., one or more of cache, SRAM, DRAM, zero capacitor RAM, Twin Transistor RAM, eDRAM, EDO RAM,

DDR RAM, EEPROM, NRAM, RRAM, SONOS, PRAM, etc.) or any one or more of the above already described possible realizations of machine-readable media. The electronic device **800** can also include a bus **810** (e.g., PCI, ISA, PCI-Express, HyperTransport®, InfiniBand®, NuBus, AHB, AXI, etc.), and network interfaces **804** can include wire-based interfaces (e.g., an Ethernet interface, a powerline communication interface, etc.). The electronic device **800** includes a communication interface **808** for network communications. The communication interface **808** can include at least one of a wireless network interface (e.g., a WLAN interface, a Bluetooth interface, a WiMAX interface, a ZigBee interface, a Wireless USB interface, etc.). In some implementations, the electronic device **800** may support multiple network interfaces—each of which is configured to couple the electronic device **800** to a different communication network.

(81) The memory unit **806** can embody functionality to implement embodiments described in FIGS. 1-6 above. In one embodiment, the memory unit **806** can include one or more of functionalities for using unified identity services in a multi-tenant architecture system. Any one of these functionalities may be partially (or entirely) implemented in hardware and/or on the processor unit **802**. For example, some functionality may be implemented with an application specific integrated circuit, in logic implemented in the processor unit **802**, in a co-processor on a peripheral device or card, etc. Further, realizations may include fewer or additional components not illustrated in FIG. 8 (e.g., video cards, audio cards, additional network interfaces, peripheral devices, etc.). The processor unit **802**, memory unit **806**, the network interfaces **804**, and the communication interface **808** are coupled to the bus **810**. Although illustrated as being coupled to the bus **810**, the memory unit **806** may be coupled to the processor unit **802**.

(82) While the embodiments are described with reference to various implementations and exploitations, it will be understood that these embodiments are illustrative and that the scope of the present disclosure is not limited to them. In general, techniques for using unified identity services in a multi-tenant architecture system as described herein may be implemented with facilities consistent with any hardware system or hardware systems. Many variations, modifications, additions, and improvements are possible.

(83) Plural instances may be provided for components, operations or structures described herein as a single instance. Finally, boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the present disclosure. In general, structures and functionality presented as separate components in the exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements may fall within the scope of the present disclosure.

Claims

1. A method comprising: receiving, at a multi-tenant platform, a request from a first entity of a first tenant of the multi-tenant platform to access a second entity, among multiple second-tenant entities of a second tenant of the multi-tenant platform, to process a transaction between the first entity and the second entity, the request including authentication information usable to authenticate the transaction with the first tenant, the first tenant comprising a first identity provider for the first entity, the second tenant comprising a second identity provider for the second entity and one or more others of the multiple second-tenant entities; generating, based at least on the authentication information and by a multi-tenant identity provider of the multi-tenant platform, an access token that includes information usable to authenticate the transaction with the first entity and the second entity, wherein the multi-tenant platform is configured for accessing, via the multi-tenant identity provider, a portion of entity data associated with one or more of the first entity and the second

entity; exchanging, by the multi-tenant platform, the access token for a tenant token for authenticating the transaction with the second tenant, the tenant token useable to authenticate the transaction with the second tenant, the tenant token being in a format of the second tenant, wherein the exchanging includes sending the access token to the second identity provider and receiving the tenant token from the second identity provider; and communicating, by the multi-tenant platform, the tenant token with the second tenant to perform the transaction.

2. The method of claim 1, wherein the first entity is a customer whose identity is managed by the first identity provider of the first tenant, wherein the second entity is a first merchant whose identity is managed by the second identity provider of the second tenant, wherein the others of the multiple second-tenant entities include one or more additional merchants, and wherein the transaction is between the customer of the first tenant and the first merchant of the second tenant.

3. The method of claim 1, wherein the request includes a first access token that is usable to authenticate the transaction with the first tenant and is in a first format used by the first tenant.

4. The method of claim 3, wherein generating the access token that includes information usable to authenticate the transaction with the first entity and the second entity includes using an internal token corresponding to the first access token.

5. The method of claim 1, wherein a payload of the access token includes: first tenant payload information including a first indication of the first entity within the first tenant; and second tenant payload information including a second indication of the first entity within the second tenant.

6. The method of claim 1, wherein the tenant token is generated using a second identity provider model at the multi-tenant platform; and wherein the second identity provider model corresponds to the second tenant and models how the tenant token is usable to authenticate the transaction with the second tenant.

7. The method of claim 1, wherein a first identity provider model includes a first indication of the first entity, wherein the first entity is an end-user; wherein a second identity provider model includes a second indication of the first entity; and wherein the first identity provider model and second identity provider model are represented within a hierarchical data structure, at the multi-tenant platform, that links the first indication and second indication together.

8. The method of claim 1, wherein the first entity is an end user; and wherein the generating the access token is in response to determining that the end user is authorized to access services of the first tenant and access services of the second tenant.

9. The method of claim 1, wherein generating the access token includes using an internal token representing access to the first tenant.

10. A multi-tenant platform that manages multiple tenants, the multi-tenant platform configured to: access a request from a first entity of a first tenant of the multi-tenant platform, the request to process a transaction between the first entity and a second entity, among multiple second-tenant entities of a second tenant of the multi-tenant platform, the request including authentication information usable to authenticate the transaction with the first tenant, the first tenant comprising a first identity provider for the first entity, the second tenant comprising a second identity provider for the second entity and one or more others of the multiple second-tenant entities; generate, based at least on the authentication information and by a multi-tenant identity provider of the multi-tenant platform, an access token that includes information usable to authenticate the transaction with the first entity and the second entity, wherein the multi-tenant platform is configured for accessing, via the multi-tenant identity provider, a portion of entity data identifying one or more of the first entity and the second entity; obtain, by the multi-tenant platform from the second identity provider and based on the access token, a tenant token for authenticating the transaction with the second tenant, the tenant token useable to authenticate the transaction with the second tenant; and communicate, by the multi-tenant platform, the tenant token with the second tenant to perform the transaction.

11. The multi-tenant platform of claim 10, wherein the first entity is a customer whose identity is managed by the first identity provider of the first tenant, wherein the second entity is a first

merchant whose identity is managed by the second identity provider of the second tenant, wherein the others of the multiple second-tenant entities include one or more additional merchants, and wherein the transaction is between the customer of the first tenant and the first merchant of the second tenant.

12. The multi-tenant platform of claim 10, wherein the request includes a first access token that is usable to authenticate the transaction with the first tenant and is in a first format used by the first tenant.

13. The multi-tenant platform of claim 10, wherein a payload of the access token includes: first tenant payload information including a first indication of the first entity within the first tenant; and second tenant payload information including a second indication of the first entity within the second tenant.

14. The multi-tenant platform of claim 10, wherein the tenant token is generated using a second identity provider model at the multi-tenant platform; and wherein the second identity provider model corresponds to the second tenant and models how the tenant token is usable to authenticate the transaction with the second tenant.

15. The multi-tenant platform of claim 10, wherein generating the access token includes using an internal token representing access to the first tenant.

16. A non-transitory machine-readable medium having instructions stored thereon, the instructions executable to cause performance of operations comprising: accessing a request from a first entity of a first tenant of a multi-tenant platform, the request to process a transaction between the first entity and a second entity, among multiple second-tenant entities of a second tenant of the multi-tenant platform, the request including authentication information usable to authenticate the transaction with the first tenant, the first tenant comprising a first identity provider for the first entity, the second tenant comprising a second identity provider for the second entity and one or more others of the multiple second-tenant entities; obtaining, based at least on the authentication information and by a multi-tenant identity provider of the multi-tenant platform, an access token that includes information usable to authenticate the transaction with the first entity and the second entity, wherein the multi-tenant platform is configured for accessing, via the multi-tenant identity provider, a portion of entity data characterizing one or more of the first entity and the second entity; determining, by the multi-tenant platform and based on the access token, a second access token for authenticating the transaction with the second tenant, the second access token useable to authenticate the transaction with the second tenant, the second access token being in a format of the second tenant, wherein the determining includes sending the access token to the second identity provider and receiving the second access token from the second identity provider; and communicating, by the multi-tenant platform, the second access token to the second tenant to perform the transaction.

17. The non-transitory machine-readable medium of claim 16, wherein the first entity is a customer whose identity is managed by the first identity provider of the first tenant, wherein the second entity is a first merchant whose identity is managed by the second identity provider of the second tenant, wherein the others of the multiple second-tenant entities include one or more additional merchants, and wherein the transaction is between the customer of the first tenant and the first merchant of the second tenant.

18. The non-transitory machine-readable medium of claim 16, wherein the request includes a first access token that is usable to authenticate the transaction with the first tenant and is in a first format used by the first tenant.

19. The non-transitory machine-readable medium of claim 16, wherein the second access token is generated using a second identity provider model at the multi-tenant platform; and wherein the second identity provider model corresponds to the second tenant and models how the second access token is usable to authenticate the transaction with the second tenant.

20. The non-transitory machine-readable medium of claim 16, wherein the first entity is an end

user; and wherein the obtaining the access token is in response to determining that the end user is authorized to access services of the first tenant and access services of the second tenant.
